Duy Nguyen

dhnguyen@caltech.edu · github.com/DNH35

Education

California Institute of Technology (Caltech) *Sep 2022 – Jun 2026*
B.S. in Computer Science & B.S. in Information and Data Science GPA: 4.1/4.0
Relevant Coursework: Statistical Foundations of Machine Learning, Quantitative Risk and Portfolio Management, Machine Learning & Data Mining, Statistical Inference, LLMs for Reasoning, Computation Theory, Algorithms,

Technical Skills

Languages/Frameworks: Python, PyTorch, TensorFlow, LangChain, LangGraph, Hugging Face, CUDA, C/C++, R, Java
Statistical & ML Methods: Generative modeling (diffusion, flow matching), time-series modeling, self-supervised learning, Monte Carlo Simulation, SHAP
Tools: Linux, FAISS, Snowflake, Git

Professional Experience

Google — Software Engineer *Sep 2026 – Present*
Trusted Partner Cloud Infrastructure
- Building change-management and compatibility infrastructure for deploying Google services across isolated cloud environments

NVIDIA — AI Applied Research Intern
USD Smart-Materials *Jun 2025 – Sep 2025*
- Built a generative AI pipeline (**SDXL, FLUX diffusion/flow-matching models + ControlNet**) that produced photorealistic renderings, significantly boosting a Vision-Language Model's material-prediction accuracy on CAD inputs
- Proposed and prototyped a video-based approach using **NVIDIA's COSMOS** model to produce temporally coherent, multi-view-consistent outputs for downstream VLM analysis

Omniverse Documents Retrieval Engine *Sep 2024 – Dec 2024*
- Built an LLM-based retrieval assistant for technical and project-document Q&A, preprocessing and indexing three modalities—text, tables, and images with **metadata-aware chunking, embeddings, and vector search**.
- Returned citation-grounded answers with source traceability, reducing manual search across distributed Omniverse documentation and accelerating engineering knowledge discovery.

USD Code *Jun 2024 – Dec 2024*
- Built a synthetic data-generation pipeline (**LangChain, GPT-4, LLaMA3**) using a persona-driven methodology, expanding NVIDIA's proprietary LLM training set **10x** and lifting eval accuracy **10%**
- Improved answer quality by building a RAG pipeline, fine-tuning LLaMA3, and integrating a FAISS vector database

Stratus Data — AI Applied Research Intern *Mar 2025 – Jun 2026*
- Architected a multi-agent data-analysis system (**LangGraph/MCP**) where a supervisor LLM delegates to specialized agents, including a Cortex Analyst agent querying a Snowflake database directly from natural language, to automate business-insight discovery

Google — STEP Intern *Jun 2023 – Sep 2023*
- Fixed bugs and extended a server-side Google Play debugging feature, giving developers visibility into app, item/non-item card, and Play Store stream diagnostics

Research Experience

Caltech — Undergraduate Researcher *Jun 2024 – Jun 2026*
Anima AI + Science Lab
- Developed a resolution-invariant time-series framework (Flow Matching + STFT/Superlet Transform) supporting multi-level supervised learning, outperforming masked autoencoding and Fourier Neural Operator baselines on neural decoding, sleep-stage classification, and activity recognition **(accepted: NeurIPS 2025 workshop, TMLR)**

Yisong Yue Lab & Eva Dyer Lab — Caltech + Georgia Tech
- Designed a novel positional encoding for self-supervised, masked-autoencoding transformers on 1D neural time-series data, removing reliance on 3D electrode locations; boosted downstream classification/decoding **15%** over supervised baselines **(accepted: NeurIPS 2025 workshop, Spotlight)**

Projects

FGNO Signal Research for Equity Alpha
- Pretrained an FGNO Transformer backbone via self-supervised flow matching on STFT spectrograms of OHLCV equity data, probed frozen representations across 8 downstream regressors and a 36-way (layer, flow-time) grid to predict 5-day forward returns
- Identified an apparently profitable mean-variance-optimized long/short signal (pooled Sharpe +2.07), then rejected it after diagnosing failure under Newey-West/bootstrap significance correction for overlapping-label autocorrelation and significant losses under walk-forward re-validation

Automated Anomaly Detection in Radiograph Images
- Trained multi-class, multi-label classifiers on approximately 50K radiographs spanning nine findings using ViT, ResNet, DenseNet, and transfer learning on an HPC cluster
- Addressed severe class imbalance with label-aware augmentation, weighted objectives, stratified evaluation, and systematic model comparison